Research Schedule and Gantt Chart

Project Title: AI and Data Analytics for Enhancing Agricultural FinTech Platforms: Enabling Smart Credit and Insurance Services for Ethiopian Farmers

Timeline: 12 months

1. Literature Review & Project Design (Month 1–2)
2. Milestones/Deliverables: Comprehensive literature review, refined research questions, and methodology.
3. Risks: Limited access to up-to-date articles.
4. Contingency Plans: Use open-access journals, request papers from authors, and utilize institutional databases.
5. Data Collection & Preparation (Month 2–4)
6. Milestones/Deliverables: Surveys/interviews with farmers and FinTech providers; secondary datasets collected; cleaned and preprocessed datasets.
7. Risks: Low survey response rate; incomplete or inconsistent secondary data.
8. Contingency Plans: Increase sample size, collaborate with cooperatives and institutions, use imputation for missing data.
9. Exploratory Data Analysis (EDA) (Month 4–5)
10. Milestones/Deliverables: Identification of key patterns and predictors; visualizations and preliminary insights.
11. Risks: Unexpected data biases or gaps.
12. Contingency Plans: Adjust feature selection, supplement with additional data, consult experts.
13. Machine Learning Model Development (Month 5–8)
14. Milestones/Deliverables: Predictive models (Random Forest, Gradient Boosting, Neural Networks); optimized and validated; performance metrics evaluated.
15. Risks: Model overfitting or poor accuracy.
16. Contingency Plans: Use cross-validation, hyperparameter tuning, ensemble methods, and alternative algorithms.
17. Integration into FinTech Platform Prototype (Month 8–10)
18. Milestones/Deliverables: Prototype system embedding models; user-friendly and scalable interface.
19. Risks: Technical integration challenges; low usability.
20. Contingency Plans: Iterative testing and redesign; gather user feedback early; simplify interface.

21. Validation & Impact Assessment (Month 10–11)

22. Milestones/Deliverables: Pilot testing with farmers and institutions; data collection on impact; comparative analysis of AI vs. traditional systems.
23. Risks: Limited participation or unreliable pilot data.
24. Contingency Plans: Engage multiple sites; adjust methods; triangulate with secondary data.

25. Analysis, Synthesis & Reporting (Month 11–12)

26. Milestones/Deliverables: Final data analysis; documentation of findings; recommendations; submission of thesis/report.
27. Risks: Delays in writing or analysis.
28. Contingency Plans: Set interim writing targets; seek supervisor feedback; allocate buffer time.

Gantt Chart (Visual Timeline)

Task	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Literature Review & Project Design	■	■										
Data Collection & Preparation		■	■	■								
Exploratory Data Analysis (EDA)				■	■							
Machine Learning Model Development					■	■	■	■				
Prototype Integration								■	■			
Validation & Impact Assessment										■	■	
Analysis, Synthesis & Reporting											■	■